CHAPTER 22 · DERIVATIONS

SOUND WAVES

2 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 BEAT FREQUENCY

SYLLABUS 22.4(B)

RESULT $f = f_1 - f_2$

1. At a point: $y_1 = A \sin 2\pi f_1 t$ and $y_2 = A \sin 2\pi f_2 t$.
2. $y = y_1 + y_2 = 2A \cos(2\pi \frac{f_1 - f_2}{2} t) \sin(2\pi \frac{f_1 + f_2}{2} t)$.
3. The sound has the average frequency $\frac{f_1 + f_2}{2}$ with a slowly varying amplitude $2A \cos(\pi(f_1 - f_2)t)$.
4. Loudness is greatest whenever the amplitude is $\pm 2A$, which happens $(f_1 - f_2)$ times per second: **beat frequency $f = f_1 - f_2$**.

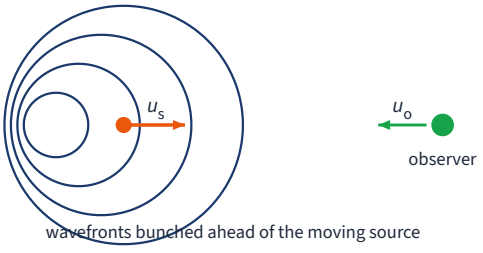
The syllabus notes that this derivation is required.

2 DOPPLER EFFECT

SYLLABUS 22.5(B)

RESULT $f' = \frac{f(v \pm u_o)}{v \mp u_s}$

1. Source of frequency f; speed of sound v.
2. **Moving source** (speed u_s towards the observer): in one period the source moves $\frac{u_s}{f}$, so the wavelength in front is shortened to $\lambda' = \frac{v - u_s}{f}$.
3. **Moving observer** (speed u_o towards the source): the waves pass the observer at relative speed $v + u_o$.
4. $f' = \frac{\text{relative speed}}{\lambda'} = \mathbf{\frac{f(v + u_o)}{v - u_s}}$ when approaching.
5. When moving apart, change the signs: $f' = \frac{f(v - u_o)}{v + u_s}$.



wavefronts bunched ahead of the moving source

Source and observer move along the same line.